

PBL PROJECT

On

**“INTELLIGENT SNAKE: ENHANCING
GAMEPLAY WITH OPENCV AI”**

Submitted to CMR ENGINEERING COLLEGE

In Partial Fulfillment of the requirements for the Award of Degree of

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**

Submitted

By

B. KISHORE KUMAR REDDY	(238R1A05Y7)
RAJ KUMAR	(238R1A05BJ)
N. MANOJ KUMAR	(248R5A0533)
P. HARSHAVARDHAN	(248R5A0535)

Under the Estimated guidance of

Mr. S. Kiran Kumar

Assistant Professor, Department of CSE



**Department of Computer Science & Engineering
CMR ENGINEERING COLLEGE**

UGC AUTONOMOUS

*(Accredited by NBA, Approved by AICTE, NEW DELHI, Affiliated to JNTU, Hyderabad)
Kandlakoya, Medchal Road, R.R. Dist. Hyderabad-501401)*

2025-2026

CMR ENGINEERING COLLEGE
UGC AUTONOMOUS

(Accredited by NBA, Approved by AICTE, NEW DELHI, Affiliated to JNTU, Hyderabad)

Kandlakoya, Medchal Road, R.R Dist, Hyderabad-501401)

Department of Computer Science & Engineering



CERTIFICATE

This is to certify that the project based learning entitled **“INTELLIGENT SNAKE: ENHANCING GAMEPLAY WITH OPENCV AI”** is a bona fide work carried out by

B. KISHORE KUMAR REDDY (238R1A05Y7)	
RAJ KUMAR	(238R1A05BJ)
N. MANOJ KUMAR	(248R5A0533)
P. HARSHAVARDHAN	(248R5A0535)

In partial fulfillment of the requirement for the award of the degree of **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE AND ENGINEERING** from CMR Engineering College, Hyderabad, under our guidance and supervision. The results presented in this project based learning have been verified and are found to be satisfactory. The results embodied in this project based learning have not been submitted to any other university for the award of any other degree or diploma.

Internal Guide

Mr. S. Kiran Kumar

Assistant Professor
Department of CSE
CMREC, Hyderabad

Head of the Department

Dr. Sheo Kumar

Professor & H.O.D
Department of CSE
CMREC, Hyderabad

DECLARATION

This is to certify that the work reported in the present project based learning entitled“ **INTELLIGENT SNAKE: ENHANCING GAMEPLAY WITH OPENCV AI**” is a record of bonafide work done by us in the Department of Computer Science and Engineering, CMR Engineering College, Hyderabad. The reports are basedon the project based learning work done entirely by us and not copied from anyother source. We submit our project based learning for further development by anyinterested students who share similar interests to improve the project based learning in the future.The results embodied in this project based learning report have notbeen submitted to any other University or Institute for the award of any degree or diploma to the best of our knowledge and belief.

B. KISHORE KUMAR REDDY	(238R1A05Y7)
RAJ KUMAR	(238R1A05BJ)
N. MANOJ KUMAR	(248R5A0533)
P. HARSHAVARDHAN	(248R5A0535)

ACKNOWLEDGMENT

We are extremely grateful to **Dr.A.Srinivasula Reddy**,Principal and **Dr.Sheo Kumar**, HOD, **Department of CSE, CMR Engineering College** for their constant support.We are extremely thankful to **Mr. S. Kiran Kumar**, Assistant Professor, Internal Guide ,Department of CSE, for his constant guidance, encouragement and moral support through out the project based learning.We will be failing in duty if I do not acknowledge with grateful thanks to the authors of the references and other literatures referred in this Project based learning.We express my thanks to all staff members and friends for all the help and co-ordination extended in bringing out this project based learning successfully in time. Finally, we are very much thankful to my parents who guided me for every step.

B. KISHORE KUMAR REDDY	(238R1A05Y7)
RAJ KUMAR	(238R1A05BJ)
N. MANOJ KUMAR	(248R5A0533)
P. HARSHAVARDHAN	(248R5A0535)

CONTENTS

TOPIC

1.ABSTRACT	i
2.INTRODUCTION	1
3.ALGORITHM	5
4.REQUIREMENTS	10
5.IMPLEMENTATION	12
6.OUTPUT SCREENS	16
7.CONCLUSION	18
8.REFERENCES	20

ABSTRACT

This project presents an interactive Snake game controlled through hand movements, utilizing Python and OpenCV for real-time hand tracking. The game leverages the cvzone library to facilitate image processing and overlay functionalities, enhancing user experience. The game is designed to be played using motion gestures captured through a webcam, with the help of the OpenCV library in Python. The game's objective remains the same: The player controls a snake that moves around the screen and eats food to grow longer while avoiding collisions with walls or the snake's own body. The study involved implementing the game using Pygame and using OpenCV to capture motion gestures, including tracking the user's hand movement and mapping it to game controls. User studies showed that the game with motion gesture control is comparable in accuracy to traditional keyboard controls but offers a more enjoyable gaming experience. This study demonstrates the potential of using computer vision and natural user interfaces for interactive and immersive gaming experiences.

INTRODUCTION

The **Snake Game using a Hand Recognition System** is an innovative blend of classic gaming and modern computer vision technology. This system allows players to control the snake using real-time hand gestures instead of traditional input devices like keyboards or touchscreens. By integrating a webcam with advanced hand recognition algorithms, the game interprets the player's hand movements and translates them into directional controls, making the gaming experience more engaging and immersive.

This application leverages OpenCV or similar libraries for hand tracking, ensuring precise and responsive gameplay. The use of hand gestures not only introduces a unique way to interact with the game but also demonstrates the potential of computer vision in enhancing user experiences. This system showcases the fusion of entertainment and technology, paving the way for more intuitive and accessible gaming interfaces in the future.

You can see me playing my all-time favorite game, which I've been playing since I was a kid, in this project. Snake Game is the name of the game. Gestures are used to manipulate the snake's position on the screen. It's your responsibility to keep the snake from hitting the wall boundaries because if it does, the game is finished. For guiding the snake, I utilized a green-colored item as a pointer.

This is a game in which the user manipulates a object in front of a webcam to create a trail that follows the object's image and resembles the shape of a snake. The snake's size keeps growing, and the game becomes increasingly tough to play.

NumPy and OpenCV are the libraries utilized. This project is classified as part of the field of gesture recognition, which is rising in popularity since it makes it very simple for users to complete specific activities. Gesture recognition has

slowly but steadily gained a foothold in the gaming business and is now expanding into other areas. It enables a far better user experience and dynamic and user-friendly interfaces when combined with computer vision.

The Snake Game using a hand recognition system is an innovative approach to merging classic gaming with modern computer vision technology. Instead of relying on traditional controls like keyboards or joysticks, this system allows players to control the snake's movement through hand gestures captured via a webcam. Utilizing advanced hand-tracking algorithms from libraries like OpenCV and cvzone, the system detects and processes hand movements in real time, making the game interactive and engaging. This hands-free mechanism not only adds a unique dimension to the gameplay but also demonstrates the practical application of gesture recognition in entertainment and technology.

This project emphasizes the potential of gesture recognition in creating intuitive, contactless interaction models. By using real-time tracking and processing algorithms, the game achieves high accuracy and responsiveness, making it engaging for players of all ages. The hands-free gameplay not only introduces a unique way of interacting with the digital environment but also highlights the broader applications of hand-tracking technology in fields such as gaming, education, and accessibility solutions. This fusion of technology and entertainment creates a fun, interactive experience while demonstrating the capabilities of advanced computer vision techniques.

Importance of Snake game a hand gesture recognition:

Integrating hand gesture recognition into the Snake game brings a range of benefits that enhance user experience, accessibility, and the overall gaming experience. Here's a breakdown of its significance:

1. Enhanced User Experience

Hand gesture recognition offers a more immersive and interactive gaming experience. Instead of using traditional input methods such as keyboard or mouse, players can control the game using natural hand movements. This provides a more intuitive and engaging way to interact with the game, making it feel more realistic and fun.

- **Intuitive Controls:** Players control the Snake using their hands, which is much more natural compared to using keyboard keys. The game reacts to the movement of the hands in real-time, offering a fluid and intuitive experience.
- **Increased Engagement:** Hand gesture recognition makes the game feel more dynamic and responsive, encouraging players to stay engaged longer. The physical aspect of using hand gestures makes it more interactive, keeping players actively involved.

2. Accessibility and Inclusivity

Hand gesture-based control systems can be particularly beneficial for people with disabilities or those who find traditional gaming inputs difficult to use. This gesture-based control removes barriers, making gaming more inclusive for a wider range of people.

- **Alternative to Mouse/Keyboard:** Individuals with motor disabilities or those unable to use a keyboard or mouse can use simple hand gestures to control the game. This opens up gaming to a broader audience who may otherwise find it difficult to engage with traditional gaming methods.
- **Natural Interaction:** Gesture control is intuitive and doesn't require special training. People of all ages and backgrounds can quickly learn to play using their hand gestures.

3. Physical Exercise and Health Benefits

Using hand gestures for controlling the Snake game promotes physical activity, which is increasingly important in today's sedentary world. This type of gaming system encourages players to move their hands and arms, engaging them in mild exercise during gameplay.

- **Promotes Hand-Eye Coordination:** Playing with hand gestures requires players to visually track the game and make corresponding hand movements. This helps improve hand-eye coordination and spatial awareness.
- **Encourages Physical Activity:** Prolonged gaming sessions using hand gestures can stimulate physical movements, thus reducing the negative impacts of extended sitting and promoting movement and exercise.

ALGORITHM

1. Initialization

1. Setup Camera:

- Capture video feed from the webcam using `cv2.VideoCapture`.
- Set the resolution of the video to 1280x720.

2. Hand Detection:

- Utilize `cvzone.HandTrackingModule` to detect and track hands in real-time with a confidence level of 0.8.
- Configure the module to track only one hand.

3. Game Setup:

- Create a `SnakeGameClass` to handle the game logic.
- Initialize variables: snake position, allowed length, score, and food position.
- Load the food image (`Donut.png`) for overlay.

2. Snake Movement

1. Capture Frame:

- Read a frame from the webcam and flip it horizontally for a mirrored view.

2. Detect Hand:

- Identify hand landmarks detected.
- Extract the position of the index finger's tip (`lmList[8][0:2]`) as the snake's current head position.

3. Track Snake:

- Append the current head position to the list of snake points.